

Description of Lab Work nº 3	
Course	Real-Time systems / <i>Sistemas de Tempo Real</i>
Year	2022/2023
Aim	Development of real-time applications using graphical languages
Classes	2 classes x 3 hours + 6 extra hours (outside classes)
Delivery Date	17/12/2022 (no penalty until 19/12/2022)
Concrete Objectives: - Utilization of graphical languages to model real-time systems - Modeling of concurrent/simultaneous behaviors - Modeling with Petri Nets - Application of PN modelling in a real problem. - Implement the real problem requirements. - Fill in a questionnaire and submit the work (according to the procedures) in Moodle.	

Annex 1 – Description of Work

The purpose of this work is to solve problems with real-time constraints using the modeling approaches based on graphical languages, namely, Petri Nets (PN) diagrams. This work is composed of two parts, namely:

1. Model and simulate PNs of several types, learn their behavior, and recognize fundamental properties (as provided in theoretical lessons).
2. Apply the PN modeling framework in the solution of a real problem.

1st Part - Use the HPSim tool for modeling PN and study their properties

In the first part of this work, we have the opportunity to simulate the behavior of Petri-Nets during their life cycle using HPSim software (Figure 1). After starting a simulation, we can visualize the enabling and firing of a PN' transitions and the tokens in each place. The work comprises modeling distinct PN types, which also exhibit fundamental PN properties (conservative, secure, etc.).

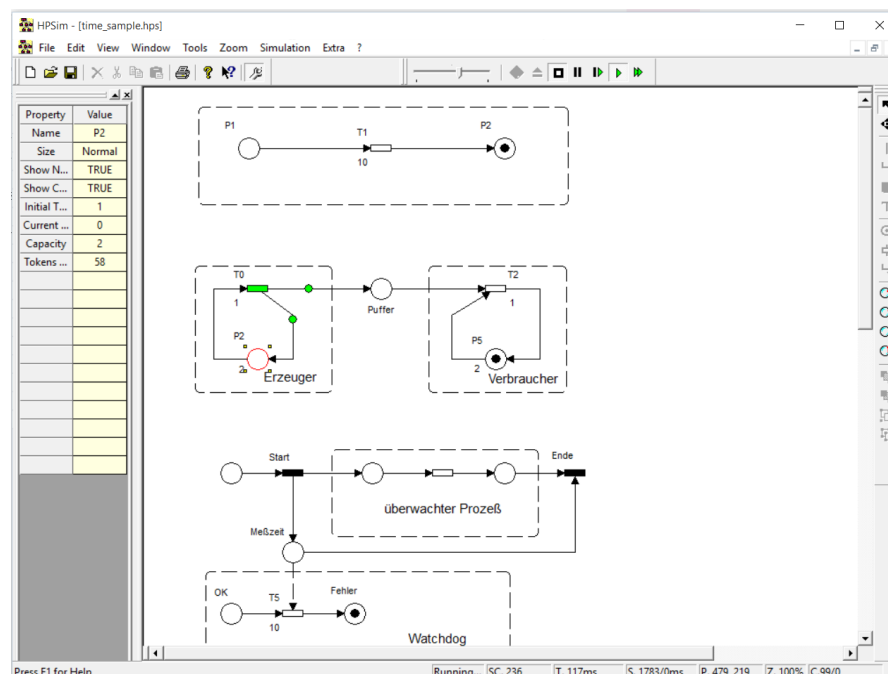


Figure 1 – The HPSim simulation tool.

2nd Part – Control a car wash station with Petri-Nets

The lab work problem consists of the design of a PN model and implementation of a program to control the car wash station illustrated in Figure 2. In Figure 3 it is represented the car wash station simulator and the corresponding sensors.

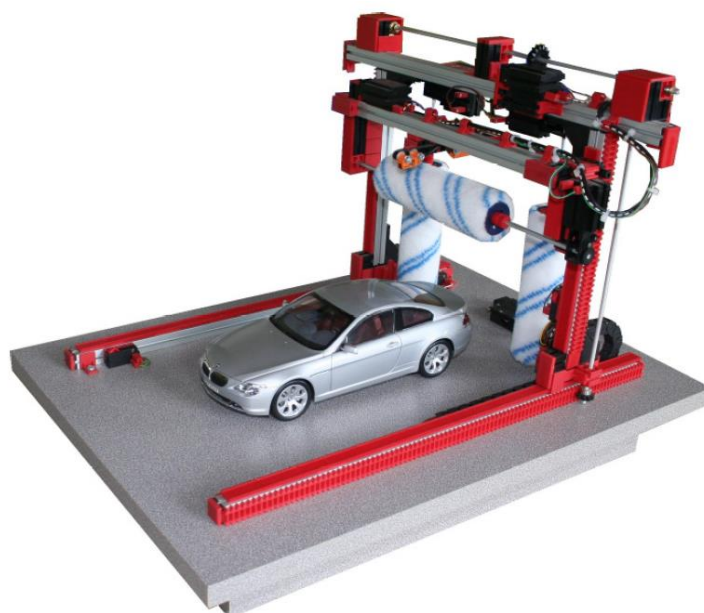


Figure 2 – The car wash station kit.

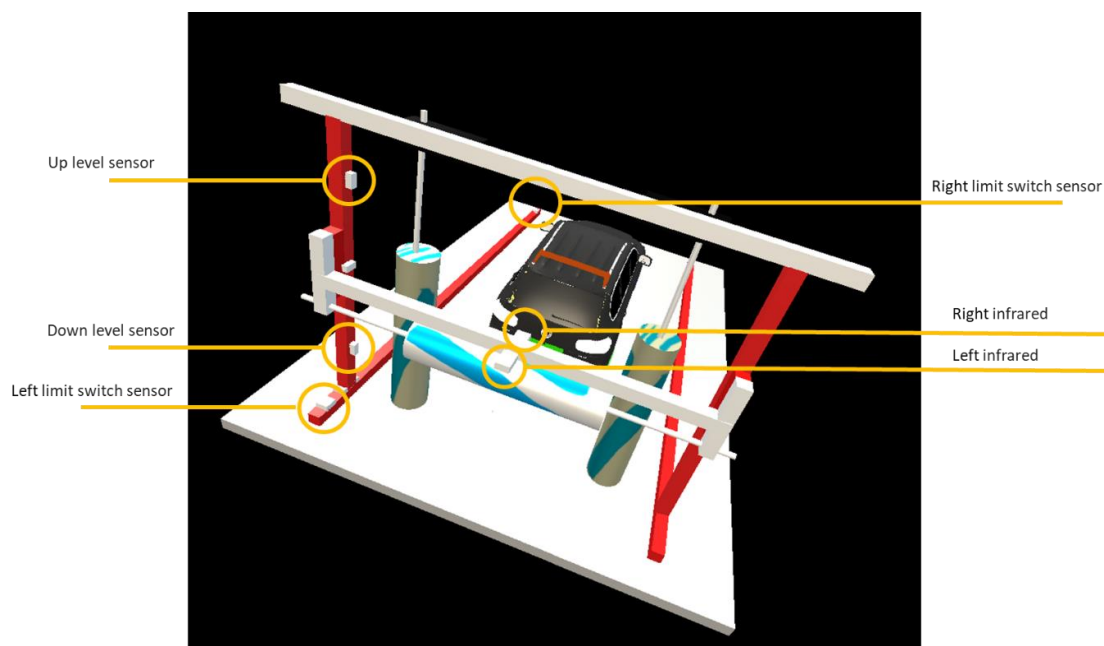


Figure 3 – The car wash simulator.

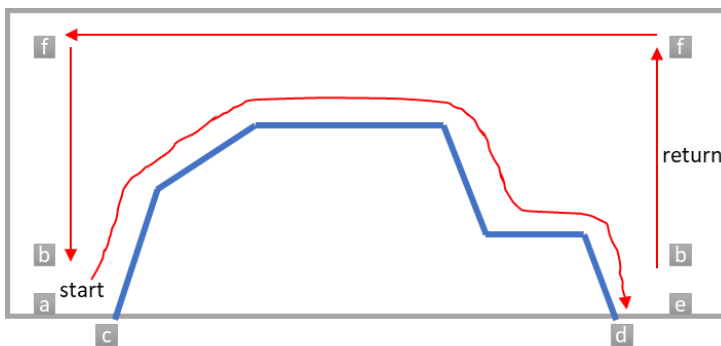
The execution of this work should be subject to the following steps:

- Study the requirements of the problem (partially defined below).
- Model the problem into a Petri net diagram.

The carwash management program must provide the following services to the costumers:

- Simple wash program (5 euros)
- Professional wash program (20 euros)
- Mechanism for emergency stop and resume

The **simple wash** program must follow the movements pattern described in Figure 4. **Only horizontal brush** is rotating during the start movement and **only the vertical brushes are rotating during the return to the original position.**



- a – left limit switch sensor
- b – down level sensor
- c – left edge of the car
- d – right edge of the car
- e – right limit switch sensor
- f – up level sensor

Figure 4 – The simple wash pattern

The **professional wash** program must follow the movements pattern described in Figure 5. This pattern comprises several steps described as follow:

Take Figure 5-(a) as illustration

- (1) movement until reaching the rear of car (brushes off).
- (2) movement with only vertical brushes on until the right edge of the car. While the washing mechanism is moving, the left/right edges of the car is reached when down level sensor activates, and infrared sensors are off.
- (3) movement with only vertical brushes on, until the right edge of the car (see step 2 for identifying the edge of the car).

Take Figure 5-(b) as illustration

- (4) movement with only horizontal brush on, until the right edge of the car. While the washing mechanism is moving, the left/right edges of the car is reached when down level sensor activates, and infrared sensors are off.
- (5) movement with horizontal and vertical brushes on until the left edge of the car.
- (6) movement to home position with all brushes off.

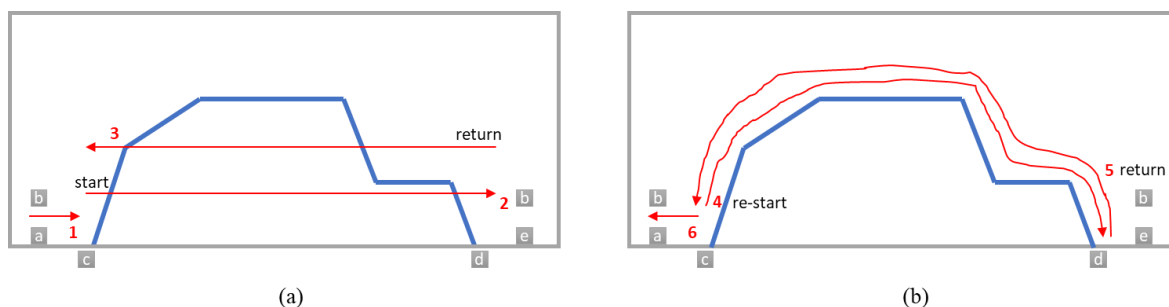


Figure 5 – The professional wash pattern; (a) vertical brush works - forward movement / vertical brushes work - return; (b) horizontal brush work - forward movement / horizontal + vertical brushes work - return.

Use the modified HPSimDLL tool (Figure 6) for modeling a PN diagram that can control the car wash station. The software was modified to include a button named DLL, which allows attaching a DLL to the PN model.

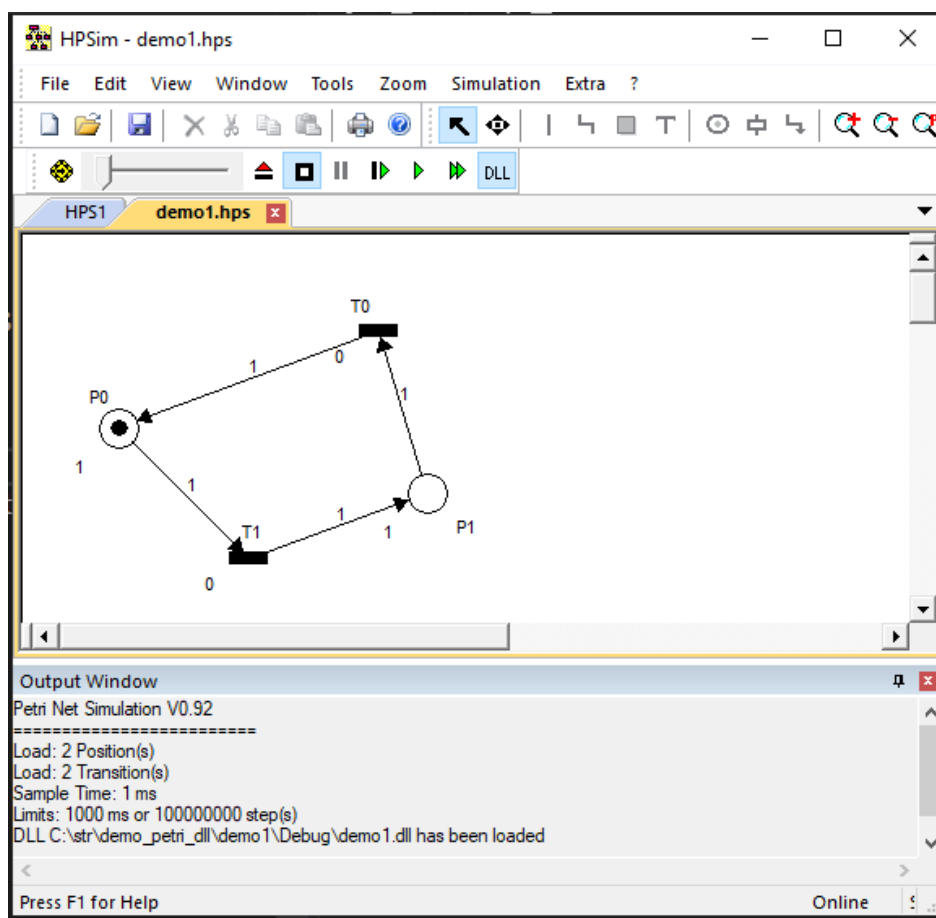


Figure 6 – Modified PN simulation tool

The requirements that need to be considered during these two parts are in the table below.

Req.	Description
	<u>Practice on Petri Net models, using HPSim. Take an image of each network and add it to the report, STR_TP3_Report.</u>
R1	Model and simulate a secure PN; PNs from theoretical classes can be used in this part.
R2	Model and simulate a conservative PN
R3	Model and simulate a PN with capacity places and weighted arcs
R4	Model and simulate a PN which starts working and enters in a deadlock state after a few iterations
	Model a PN to control the carwash station:
R5	Simple wash program
R6	Professional wash program
R7	Emergency stop and resume button
R8	Describe your lab work in the report (STR_TP3_Report).
R9	Identify, from previous experiments, the different characteristics of your proposed PN for your lab work (Example, list from slide 14, STR_Lab3_1) and describe it in the report (STR_TP3_Report).

Classes plan for the Lab Work 3

Class 1:

Introduction to *HPSim* (1st part) for training with Petri nets; Start developing the lab work (2nd part). Creation of DLL and its utilization in *HPSim* (modified version).

Class 2:

Finish developing the lab work (2nd part). Control of the car wash station. Wash services development. Use GUI for user interaction.

Professors:

- | | | |
|-------------------------|--|-----------------------|
| – Luis Camarinha-Matos, | cam@uninova.pt | (theoretical classes) |
| – Ana Inês Oliveira, | aio@fct.unl.pt | (laboratory classes) |
| – André Rocha, | andre.rocha@uninova.pt | (laboratory classes) |
| – Filipa Ferrada, | fam@fct.unl.pt | (laboratory classes) |
| – Francisco Marques | fam@uninova.pt | (laboratory classes) |

Lab work 3 submission & delivery date:

- The lab work must be submitted **only** through **Moodle** until **17.12.2022 – 23h:59m**. No penalty until 19.12.2022 – 23h:59m.
- The work is submitted in **zip format**, the zip name should have the student numbers of each group member separated by ‘_’, e.g.: 50234_51345_51734_50599.zip. The zip file must contain:
 - Report.
 - the *carWashServices* project (DLL), including the PN model, developed in Part 2.
- It must also be filled in the **questionnaire** available in Moodle. In this questionnaire it is requested to submit a small video (3-5 minutes) showing the work in operation.
- The project *zip* file, the questionnaire and the video are **mandatory** components of evaluation.

Evaluation criteria

It is provided (as a guiding object) a table with the project evaluation criteria. Each item will be quoted according to the approach and quality of the generated code.

Items		Quotation
Functionality	R1 - Model and simulate a secure PN	0.50
	R2 - Model and simulate a conservative PN	0.50
	R3 - Model and simulate a PN with capacity places and weighted arcs	1.00
	R4 - Model and simulate a PN which starts working and enters in a deadlock state after a few iterations	1.00
	R5 - Simple wash program	3.00
	R6 - Professional wash program	4.00
	R7 - Emergency stop and resume button	3.00
	R8 - Describe your lab work in the report (STR_TP3_Report).	2.00
	R9 - Identify, from previous experiments, the different characteristics of your proposed PN for your lab work (Example, list from slide 14, STR_Lab3_1) and describe it in the report (STR_TP3_Report).	2.00
Video		3.00
TOTAL		20